NAME:NITHIN.V

ROLL.NO:2403A52355

BATCH.NO:13(AIML)

STEP 1 — Install and Import Required Libraries

```
# This installs Hugging Face Transformers and PyTorch
!pip install transformers
!pip install torch
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.1)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transform
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->h
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->h
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer-
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.1)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
```

```
# Import necessary libraries

from transformers import pipeline   # Used to load pretrained NLP pipelines
import torch                        # Backend deep learning library

# Verify installation
print("Torch version:", torch.__version__)
print("Transformers successfully imported!")
```

```
Torch version: 2.10.0+cpu
Transformers successfully imported!
```

STEP 2 — Load Translation Model

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# English to French translator
model_name_en_fr = "Helsinki-NLP/opus-mt-en-fr"
tokenizer_en_fr = AutoTokenizer.from_pretrained(model_name_en_fr)
model_en_fr = AutoModelForSeq2SeqLM.from_pretrained(model_name_en_fr)

def translator_en_fr(text):
    inputs = tokenizer_en_fr(text, return_tensors="pt", padding=True, truncation=True)
    translated_tokens = model_en_fr.generate(**inputs, max_length=512)
    return [{'translation_text': tokenizer_en_fr.decode(translated_tokens[0], skip_special_tokens=True)}]

# English to German translator
model_name_en_de = "Helsinki-NLP/opus-mt-en-de"
tokenizer_en_de = AutoTokenizer.from_pretrained(model_name_en_de)
```

```
model_en_de = AutoModelForSeq2SeqLM.from_pretrained(model_name_en_de)

def translator_en_de(text):
    inputs = tokenizer_en_de(text, return_tensors="pt", padding=True, truncation=True)
    translated_tokens = model_en_de.generate(**inputs, max_length=512)
    return [{'translation_text': tokenizer_en_de.decode(translated_tokens[0], skip_special_tokens=True)}]

# English to Hindi translator
model_name_en_hi = "Helsinki-NLP/opus-mt-en-hi"
tokenizer_en_hi = AutoTokenizer.from_pretrained(model_name_en_hi)
model_en_hi = AutoModelForSeq2SeqLM.from_pretrained(model_name_en_hi)

def translator_en_hi(text):
    inputs = tokenizer_en_hi(text, return_tensors="pt", padding=True, truncation=True)
    translated_tokens = model_en_hi.generate(**inputs, max_length=512)
    return [{'translation_text': tokenizer_en_hi.decode(translated_tokens[0], skip_special_tokens=True)}]

print("Translation models loaded successfully")
```

```
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: pip
  warnings.warn("Recommended: pip install sacremoses.")
Loading weights: 100%|| 258/258 [00:01<00:00, 183.81it/s, Materializing param=model.shared.weight]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight
config.json: 1.33k/? [00:00<00:00, 24.3kB/s]
tokenizer_config.json: 100%|| 42.0/42.0 [00:00<00:00, 3.63kB/s]
source.spm: 100%|| 768k/768k [00:00<00:00, 12.7MB/s]
target.spm: 100%|| 797k/797k [00:00<00:00, 38.5MB/s]
vocab.json: 1.27M/? [00:00<00:00, 31.2MB/s]
pytorch_model.bin: 100%|| 298M/298M [00:03<00:00, 128MB/s]
Loading weights: 100%|| 258/258 [00:00<00:00, 637.25it/s, Materializing param=model.shared.weight]
model.safetensors: 100%|| 298M/298M [00:04<00:00, 84.7MB/s]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight
generation_config.json: 100%|| 293/293 [00:00<00:00, 12.9kB/s]
config.json: 1.39k/? [00:00<00:00, 85.0kB/s]
tokenizer_config.json: 100%|| 44.0/44.0 [00:00<00:00, 2.72kB/s]
source.spm: 100%|| 812k/812k [00:00<00:00, 29.1MB/s]
target.spm: 100%|| 1.07M/1.07M [00:00<00:00, 36.3MB/s]
vocab.json: 2.10M/? [00:00<00:00, 37.6MB/s]
pytorch_model.bin: 100%|| 306M/306M [00:02<00:00, 182MB/s]
Loading weights: 100%|| 258/258 [00:00<00:00, 499.58it/s, Materializing param=model.shared.weight]
model.safetensors: 100%|| 306M/306M [00:03<00:00, 142MB/s]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight
generation_config.json: 100%|| 293/293 [00:00<00:00, 26.6kB/s]
Translation models loaded successfully
```

STEP 3 — Prepare Input Sentences (Dataset)

```
# Create dataset sentences

sentences = [

    # Simple sentences
    "The sun rises in the east.",
    "She likes reading books.",

    # Compound sentences
    "I wanted to go for a walk, but it started raining.",
    "He studied hard and passed the exam.",

    # Complex sentences
    "Although the weather was cold, we decided to go hiking.",
    "When the meeting ended, everyone left the room.",

    # Interrogative sentences
```

```
    "What time does the train arrive?",
    "Why are you learning machine learning?",

    # Technical sentences
    "Artificial intelligence is transforming many industries.",
    "Neural networks are used in deep learning models.",

    # Conversational sentences
    "Hey, how are you doing today?",
    "Let's grab some coffee after class.",

    # News-style sentences
    "The government announced new policies to improve education.",

    # Academic text
    "Natural language processing enables machines to understand human language.",

    # Social media style
    "This new phone camera is amazing!"
]

print("Total sentences:", len(sentences))
```

```
Total sentences: 15
```

STEP 4 — Perform Translation

```
# Translate sentences and display results

print("English → French Translation\n")

for sentence in sentences:

    # Translate sentence
    result = translator_en_fr(sentence)

    # Extract translated text
    translated = result[0]['translation_text']

    # Print results
    print("Original :", sentence)
    print("French   :", translated)
    print("-"*60)
```

```
English → French Translation

Original : The sun rises in the east.
French   : Le soleil se lève à l'est.
------------------------------------------------------------
Original : She likes reading books.
French   : Elle aime lire des livres.
------------------------------------------------------------
Original : I wanted to go for a walk, but it started raining.
French   : Je voulais faire une promenade, mais il a commencé à pleuvoir.
------------------------------------------------------------
Original : He studied hard and passed the exam.
French   : Il a étudié dur et a réussi l'examen.
------------------------------------------------------------
Original : Although the weather was cold, we decided to go hiking.
French   : Bien que le temps fût froid, nous avons décidé d'aller à la randonnée.
------------------------------------------------------------
Original : When the meeting ended, everyone left the room.
French   : Quand la réunion s'est terminée, tout le monde a quitté la pièce.
------------------------------------------------------------
Original : What time does the train arrive?
French   : A quelle heure arrive le train ?
------------------------------------------------------------
Original : Why are you learning machine learning?
French   : Pourquoi apprenez-vous l'apprentissage automatique ?
------------------------------------------------------------
Original : Artificial intelligence is transforming many industries.
French   : L'intelligence artificielle transforme de nombreuses industries.
------------------------------------------------------------
Original : Neural networks are used in deep learning models.
French   : Les réseaux neuraux sont utilisés dans les modèles d'apprentissage profond.
------------------------------------------------------------
Original : Hey, how are you doing today?
French   : Comment ça va aujourd'hui ?
------------------------------------------------------------
Original : Let's grab some coffee after class.
French   : Allons prendre un café après le cours.
------------------------------------------------------------
Original : The government announced new policies to improve education.
French   : Le gouvernement a annoncé de nouvelles politiques visant à améliorer l'éducation.
------------------------------------------------------------
```

```
Original : Natural language processing enables machines to understand human language.
French   : Le traitement naturel du langage permet aux machines de comprendre le langage humain.
------------------------------------------------------------
Original : This new phone camera is amazing!
French   : Cette nouvelle caméra est incroyable !
------------------------------------------------------------
```

STEP 5 — Translate into Multiple Languages

```
#  Translate into multiple languages

translations = []

for sentence in sentences:

    fr = translator_en_fr(sentence)[0]['translation_text']
    de = translator_en_de(sentence)[0]['translation_text']
    hi = translator_en_hi(sentence)[0]['translation_text']

    translations.append({
        "English": sentence,
        "French": fr,
        "German": de,
        "Hindi": hi
    })

# Display translations
for t in translations:

    print("English :", t["English"])
    print("French  :", t["French"])
    print("German  :", t["German"])
    print("Hindi   :", t["Hindi"])
    print("="*70)
```

```
English : The sun rises in the east.
French  : Le soleil se lève à l'est.
German  : Die Sonne geht im Osten auf.
Hindi   : सूरज पूर्व में उदय होता है.
======================================================================
English : She likes reading books.
French  : Elle aime lire des livres.
German  : Sie liest gern Bücher.
Hindi   : वह किताबों को पढ़ना पसंद करती है.
======================================================================
English : I wanted to go for a walk, but it started raining.
French  : Je voulais faire une promenade, mais il a commencé à pleuvoir.
German  : Ich wollte spazieren gehen, aber es fing an zu regnen.
Hindi   : मैं एक चलने के लिए जाना चाहता था, लेकिन यह बारिश शुरू कर दिया.
======================================================================
English : He studied hard and passed the exam.
French  : Il a étudié dur et a réussi l'examen.
German  : Er hat hart studiert und die Prüfung bestanden.
Hindi   : उसने कड़ी मेहनत की और परीक्षा का सामना किया ।
======================================================================
English : Although the weather was cold, we decided to go hiking.
French  : Bien que le temps fût froid, nous avons décidé d'aller à la randonnée.
German  : Obwohl das Wetter kalt war, beschlossen wir zu wandern.
Hindi   : हालाँकि मौसम ठंडा था, फिर भी हमने खाने का फैसला किया ।
======================================================================
English : When the meeting ended, everyone left the room.
French  : Quand la réunion s'est terminée, tout le monde a quitté la pièce.
German  : Als das Treffen zu Ende war, verließen alle den Raum.
Hindi   : जब सभा खत्म हुई, तो सभी ने कमरे छोड़ दिए ।
======================================================================
English : What time does the train arrive?
French  : A quelle heure arrive le train ?
German  : Wann kommt der Zug an?
Hindi   : ट्रेन किस वक्त आती है?
======================================================================
English : Why are you learning machine learning?
French  : Pourquoi apprenez-vous l'apprentissage automatique ?
German  : Warum lernst du maschinelles Lernen?
Hindi   : आप मशीन सीखने क्यों कर रहे हैं?
======================================================================
English : Artificial intelligence is transforming many industries.
French  : L'intelligence artificielle transforme de nombreuses industries.
German  : Künstliche Intelligenz verändert viele Branchen.
Hindi   : कलापूर्ण बुद्धि अनेक उद्योगों को परिवर्तित कर रही है।
======================================================================
English : Neural networks are used in deep learning models.
French  : Les réseaux neuraux sont utilisés dans les modèles d'apprentissage profond.
German  : Neurale Netzwerke werden in Deep-Learning-Modellen eingesetzt.
Hindi   : तंत्रिका नेटवर्कों को गहरी सीखने वाले मॉडलों में इस्तेमाल किया जाता है।
======================================================================
English : Hey, how are you doing today?
```

```
French  : Comment ça va aujourd'hui ?
German  : Hey, wie geht's dir heute?
Hindi   : अरे, आप आज कैसे कर रहे हैं?
======================================================================
English : Let's grab some coffee after class.
French  : Allons prendre un café après le cours.
German  : Lass uns nach dem Unterricht einen Kaffee trinken.
```

STEP 6 — Analyze Translation Quality

```
#  Store translations for analysis

import pandas as pd

df = pd.DataFrame(translations)

# Display translation table
df
```

	English	French	German	Hindi
0	The sun rises in the east.	Le soleil se lève à l'est.	Die Sonne geht im Osten auf.	सूरज पूर्व में उदय होता है.
1	She likes reading books.	Elle aime lire des livres.	Sie liest gern Bücher.	वह किताबों को पढ़ना पसंद करती है.
2	I wanted to go for a walk, but it started rain...	Je voulais faire une promenade, mais il a comm...	Ich wollte spazieren gehen, aber es fing an zu...	मैं एक चलने के लिए जाना चाहता था, लेकिन यह बार...
3	He studied hard and passed the exam.	Il a étudié dur et a réussi l'examen.	Er hat hart studiert und die Prüfung bestanden.	उसने कड़ी मेहनत की और परीक्षा का सामना किया ।
4	Although the weather was cold, we decided to g...	Bien que le temps fût froid, nous avons décidé...	Obwohl das Wetter kalt war, beschlossen wir zu...	हालाँकि मौसम ठंडा था, फिर भी हमने खाने का फैसल...
5	When the meeting ended, everyone left the room.	Quand la réunion s'est terminée, tout le monde...	Als das Treffen zu Ende war, verließen alle de...	जब सभा खत्म हुई, तो सभी ने कमरे छोड़ दिए ।
6	What time does the train arrive?	A quelle heure arrive le train ?	Wann kommt der Zug an?	ट्रेन किस वक्त आती है?
7	Why are you learning machine learning?	Pourquoi apprenez-vous l'apprentissage automat...	Warum lernst du maschinelles Lernen?	आप मशीन सीखने क्यों कर रहे हैं?
8	Artificial intelligence is transforming many i...	L'intelligence artificielle transforme de nomb...	Künstliche Intelligenz verändert viele Branchen.	कलापूर्ण बुद्धि अनेक उद्योगों को परिवर्तित कर ...
9	Neural networks are used in deep learning models.	Les réseaux neuraux sont utilisés dans les mod...	Neurale Netzwerke werden in Deep-Learning-Mode...	तंत्रिका नेटवर्कों को गहरी सीखने वाले मॉडलों म...
10	Hey, how are you doing today?	Comment ça va aujourd'hui ?	Hey, wie geht's dir heute?	अरे, आप आज कैसे कर रहे हैं?

Next steps: Generate code with df New interactive sheet

STEP 7 — Test with Complex Sentences

```
#  Complex sentence testing

complex_sentences = [

    "If artificial intelligence continues to evolve rapidly, it may transform healthcare, education, and transportation indu

    "Although the project was difficult, the team managed to complete it successfully before the deadline.",

    "Can machine learning algorithms truly understand human emotions?",

    "Researchers are developing advanced neural networks that can generate realistic images and text."
]

for s in complex_sentences:

    fr = translator_en_fr(s)[0]['translation_text']
    de = translator_en_de(s)[0]['translation_text']
    hi = translator_en_hi(s)[0]['translation_text']

    print("English :", s)
    print("French  :", fr)
    print("German  :", de)
    print("Hindi   :", hi)
    print("="*70)
```

```
English : If artificial intelligence continues to evolve rapidly, it may transform healthcare, education, and transportatior
French  : Si l'intelligence artificielle continue d'évoluer rapidement, elle peut transformer les secteurs des soins de sant
German  : Wenn sich die künstliche Intelligenz weiter rasch entwickelt, kann sie die Gesundheits-, Bildungs- und Transportir
Hindi   : यदि कृत्रिम बुद्धि तेज़ी से बढ़ती जा रही है, तो यह स्वास्थ्य चिकित्सा, शिक्षा और यातायात उद्योग को बदल सकता है ।
======================================================================
```

```
English : Although the project was difficult, the team managed to complete it successfully before the deadline.
French  : Bien que le projet ait été difficile, l'équipe a réussi à l'achever avant la date limite.
German  : Obwohl das Projekt schwierig war, gelang es dem Team, es vor Ablauf der Frist erfolgreich abzuschließen.
Hindi   : हालाँकि यह काम बहुत मुश्किल था, फिर भी इस टीम ने मरे हुओं के आने से पहले इसे सफलतापूर्वक पूरा किया ।
====================================================================
English : Can machine learning algorithms truly understand human emotions?
French  : Les algorithmes d'apprentissage automatique peuvent-ils vraiment comprendre les émotions humaines?
German  : Können Algorithmen des maschinellen Lernens menschliche Emotionen wirklich verstehen?
Hindi   : क्या मशीन सीखने से इंसान की भावनाओं को सचमुच समझ सकता है?
====================================================================
English : Researchers are developing advanced neural networks that can generate realistic images and text.
French  : Les chercheurs développent des réseaux neuronaux avancés qui peuvent générer des images et des textes réalistes.
German  : Forscher entwickeln fortschrittliche neuronale Netzwerke, die realistische Bilder und Texte erzeugen können.
Hindi   : खोजकर्ताओं का कहना है कि दिमाग में नयी - नयी जानकारी होती है, जो सही - सही तसवीरें और पाठ तैयार कर सकती हैं ।
====================================================================
```

STEP 8 — Example Report Structure

1. Objective

To understand Seq2Seq models and perform machine translation using pretrained Transformer models.

2. Dataset Description

The dataset contains 15 sentences collected from:

news articles

social media

academic texts

daily conversations.

These sentences represent different structures such as simple, compound, complex, interrogative, and technical sentences.

3. Installation

Libraries installed:

transformers

torch

4. Translation Outputs

Include table or screenshots from the notebook.

5. Multi-language Comparison

Compare English → French, German, Hindi.

6. Analysis

Discuss translation accuracy, grammar, and fluency.

7. Conclusion

Pretrained Transformer models provide high-quality translations but may face difficulties with long sentences or cultural expressions.